Human Physiology: Breathing and Exchange of Gases

Target Examination: NEET UG (Biology — Human Physiology)

Scope: Respiratory Organs, Mechanism of Breathing, Respiratory Volumes and Capacities, Exchange of Gases, Transport of Gases, Regulation of Respiration, and Respiratory Disorders

1. Unit Context, Pioneers, and Learning Objectives

Systems Biology vs. Reductionist Approach in Physiology

Historically, the study of living organisms relied heavily on a **reductionist approach**, which applies physico-chemical concepts and techniques to surviving tissue models or cell-free systems. This approach caused an explosion of knowledge in molecular biology, making molecular physiology nearly synonymous with biochemistry and biophysics. However, neither a purely organismic approach nor a purely reductionist molecular approach reveals the complete truth about biological processes.

Systems biology demonstrates that living phenomena are **emergent properties** arising from interactions among components of a system. Regulatory networks of molecules, supramolecular assemblies, cells, tissues, organisms, populations, and communities each generate emergent properties. In human physiology, processes such as gas exchange and blood circulation are understood at cellular and molecular levels while being coordinated at the whole-organism level.

Historical Pioneer in Mammalian Physiology: Alfonso Corti (1822–1888)

Alfonso Corti, an Italian anatomist born in 1822, began his scientific career studying the cardiovascular systems of reptiles. Later, he turned his attention to the mammalian auditory system. In 1851, he published a landmark paper describing a structure located on the basilar membrane of the cochlea containing hair cells that convert sound vibrations into nerve impulses—now known as the **organ of Corti**. He died in 1888.

Learning Objectives

- Compare respiratory structures across invertebrate and vertebrate animal groups based on habitat and level of organisation.
- Distinguish between the anatomical components and functions of the conducting part and the exchange part of the human respiratory system.
- Explain the pressure gradients and muscular actions (diaphragm and intercostals) driving inspiration and expiration.
- Define and calculate respiratory volumes and pulmonary capacities measured via spirometry.
- Analyze partial pressure gradients ($\mathrm{pO_2}$ and $\mathrm{pCO_2}$) and structural layers of the alveolar-capillary diffusion membrane.
- Describe the quantitative transport mechanisms of $\mathrm{O_2}$ and $\mathrm{CO_2}$ in blood, including the oxygen dissociation curve and carbonic anhydrase activity.
- Explain neural and chemical regulation of respiratory rhythm and identify major respiratory disorders.

2. Core Concepts and Detailed Explanations

2.1 Introduction to Breathing and Respiration

Organisms utilise oxygen ($\mathrm{O_2}$) to indirectly break down simple molecules such as glucose, amino acids, and fatty acids to derive energy for cellular activities. During these catabolic reactions, carbon dioxide ($\mathrm{CO_2}$), which is harmful to cells, is released. Therefore, $\mathrm{O_2}$ must be continuously supplied to cells and $\mathrm{CO_2}$ must be

continuously removed. The process of exchanging O_2 from the atmosphere with CO_2 produced by the cells is called **breathing**, commonly known as **respiration**.

> **Five Sequential Steps of Human Respiration:**
>
> 1. **Breathing or pulmonary ventilation:** Drawing in atmospheric air and releasing CO_2-rich alveolar air.
> 2. **Diffusion across alveolar membrane:** Exchange of O_2 and CO_2 across the alveolar-capillary barrier.
> 3. **Transport of gases:** Carriage of O_2 and CO_2 by the circulating blood.
> 4. **Diffusion between blood and tissues:** Uptake of O_2 by tissues and release of CO_2 into blood.
> 5. **Cellular respiration:** Utilisation of O_2 by cells for catabolic reactions and resultant production of CO_2 .

2.2 Respiratory Organs Across the Animal Kingdom

Breathing mechanisms vary among animal groups depending primarily on their **habitats** and **levels of organisation**:

Animal Group	Respiratory Structure / Mode	Key Mechanism Details
Lower invertebrates (Sponges, Coelenterates, Flatworms)	Entire body surface	Simple diffusion of O_2 and CO_2 directly across the general body surface.
Earthworms	Moist cuticle (Cutaneous respiration)	Gas exchange through the moist outer cuticle covering the body.
Insects	Tracheal tubes (Tracheal respiration)	A network of tubes transports atmospheric air directly within the body to tissues.
Aquatic arthropods, Molluscs, Fishes	Gills (Branchial respiration)	Special vascularised structures adapted for gas exchange in aquatic habitats.
Amphibians (e.g., Frogs)	Lungs (Pulmonary) and Moist skin (Cutaneous)	Respire through vascularised bags (lungs) as well as their moist skin.
Reptiles, Birds, Mammals	Lungs (Pulmonary respiration)	Terrestrial forms using highly vascularised lungs for gas exchange.

2.3 Human Respiratory System

The human respiratory pathway follows a strict anatomical sequence from the exterior to the gas-exchange surfaces:

External nostrils (pair opening above the upper lips) → **Nasal passage** → **Nasal chamber** → **Pharynx** (a portion serves as the common passage for food and air) → **Larynx** (through the glottis) → **Trachea** → **Right and Left Primary Bronchi** → **Secondary Bronchi** → **Tertiary Bronchi** → **Initial Bronchioles** → **Terminal Bronchioles** → **Alveoli and alveolar ducts**.

- **Larynx and Epiglottis:** The larynx is a cartilaginous box that helps in sound production and is therefore called the **sound box**. During swallowing, the **glottis** can be covered by a thin elastic cartilaginous flap called the **epiglottis** to prevent food from entering the larynx.
- **Trachea and Bronchial Tree:** The trachea is a straight tube extending up to the mid-thoracic cavity, where it divides at the level of the **5th thoracic vertebra** into right and left primary bronchi.
- **Cartilaginous Support:** The trachea, primary, secondary, and tertiary bronchi, and *initial bronchioles* are supported by **incomplete cartilaginous rings** to prevent collapse.
- **Alveoli:** Each terminal bronchiole gives rise to a number of very thin, irregular-walled, and vascularised bag-like structures called **alveoli**. The branching network of bronchi, bronchioles, and alveoli comprises the **lungs**.
- **Pleura:** Humans have two lungs covered by a double-layered **pleura**, with **pleural fluid** between the layers. This fluid reduces friction on the lung surface. The outer pleural membrane is in close contact with the

thoracic lining, whereas the inner pleural membrane is in contact with the lung surface.

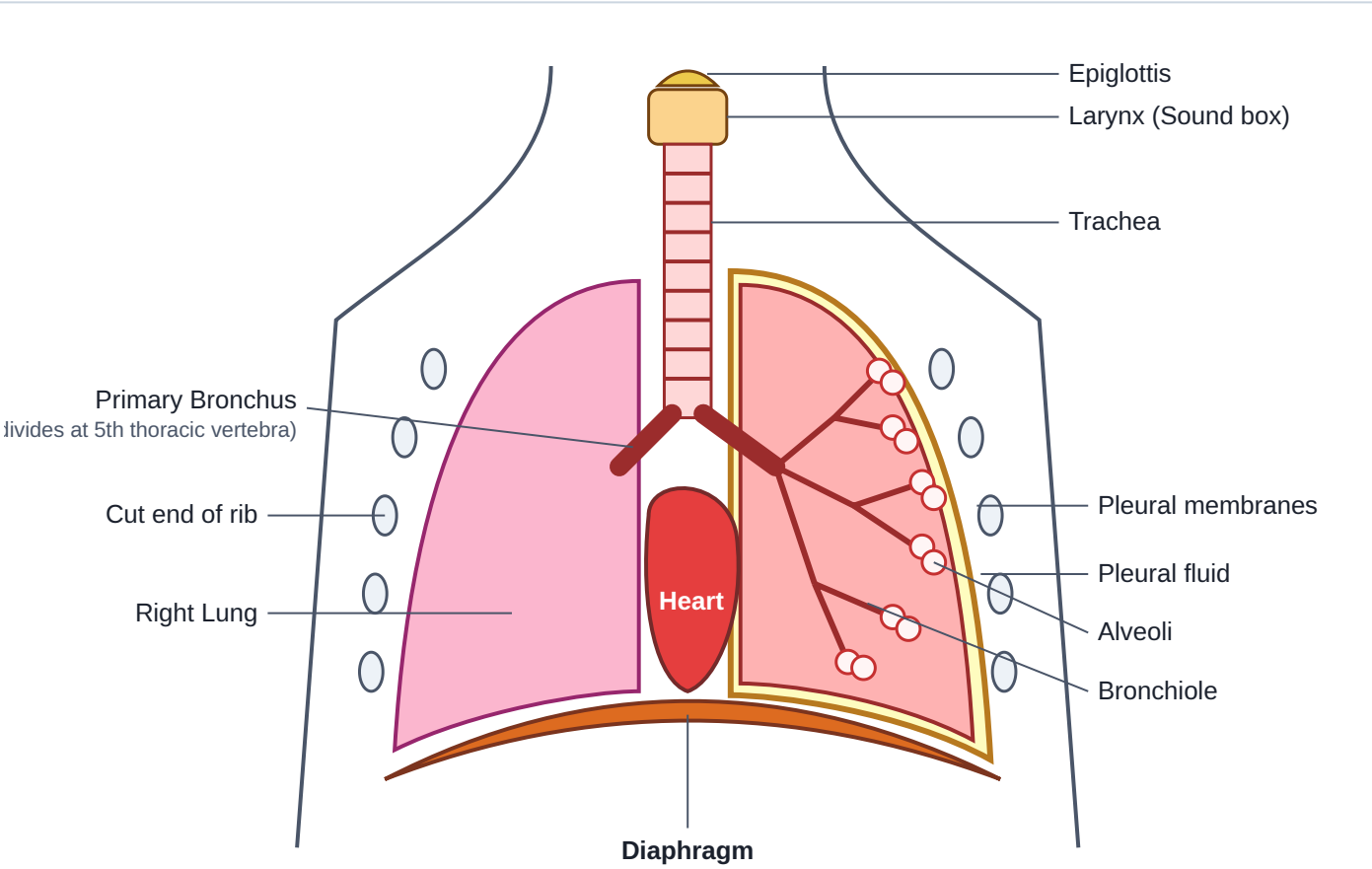
Feature	Conducting Part	Respiratory (Exchange) Part
Anatomical Extent	External nostrils up to the terminal bronchioles.	Alveoli and their ducts.
Primary Functions	1. Transports atmospheric air to the alveoli. 2. Clears air from foreign particles. 3. Humidifies the atmospheric air. 4. Brings the air to body temperature.	Site of actual diffusion of O_2 and CO_2 between blood and atmospheric air.

The Thoracic Chamber

The lungs are situated in the **thoracic chamber**, which is anatomically an **air-tight chamber**. Its boundaries are formed by:

- **Dorsally:** Vertebral column
- **Ventrally:** Sternum
- **Laterally:** Ribs
- **Lower side:** Dome-shaped diaphragm

Because we cannot directly alter pulmonary (lung) volume, this anatomical setup is essential: any change in the volume of the thoracic cavity is immediately reflected in the lung (pulmonary) cavity.



Tutorial Diagram 1: Diagrammatic view of the human respiratory system (showing sectional view of the left lung).

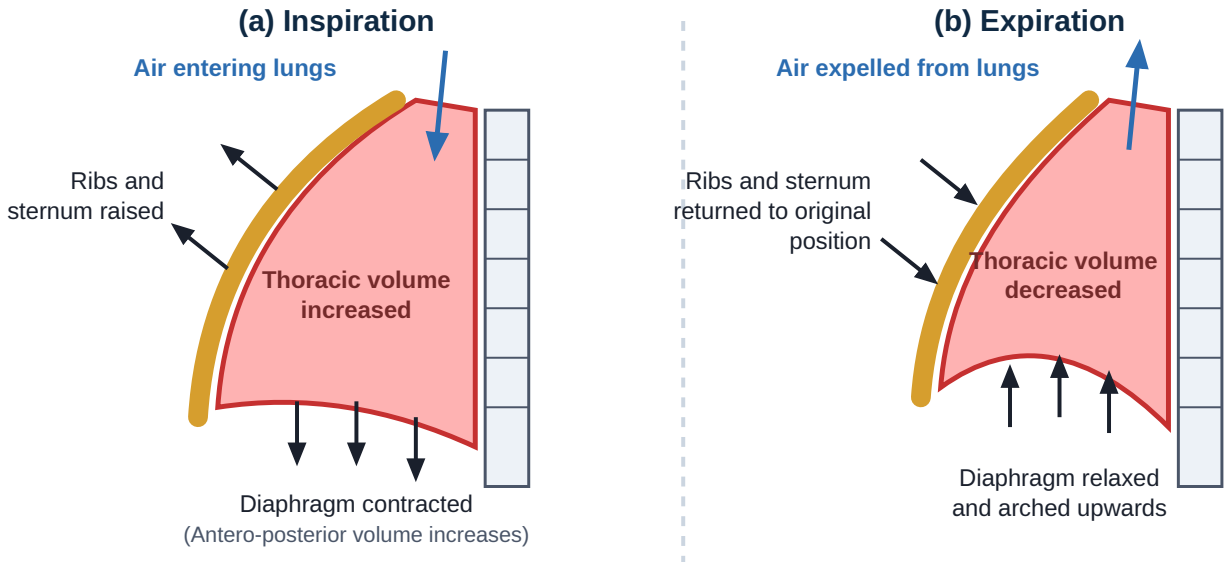
2.4 Mechanism of Breathing

Breathing involves two stages: **inspiration** (drawing atmospheric air in) and **expiration** (releasing alveolar air out). Air moves into and out of the lungs due to a **pressure gradient** generated between the lungs and the

atmosphere by the **diaphragm** and a specialised set of muscles—the **external and internal intercostals** between the ribs.

Parameter	Inspiration (Active Process)	Normal Expiration (Passive Relaxation)
Intra-pulmonary Pressure vs. Atmospheric Pressure	Intra-pulmonary pressure is **less than** atmospheric pressure (negative pressure in lungs with respect to atmosphere).	Intra-pulmonary pressure is **slightly higher than** atmospheric pressure (positive pressure with respect to atmosphere).
Diaphragm Action	**Contracts** (flattens), increasing thoracic volume in the **antero-posterior axis**.	**Relaxes** and arches upwards to its normal dome-shaped position.
External Intercostal Muscles	**Contract**, lifting up the ribs and sternum, increasing thoracic volume in the **dorso-ventral axis**.	**Relax**, returning the ribs and sternum to their original positions.
Thoracic & Pulmonary Volume	Thoracic volume increases → Pulmonary volume increases → Intra-pulmonary pressure decreases.	Thoracic volume decreases → Pulmonary volume decreases → Intra-pulmonary pressure increases.
Net Air Movement	Air moves from outside into the lungs.	Air is expelled from the lungs to the outside.

- **Forceful Breathing:** Humans have the ability to increase the strength of inspiration and expiration with the help of additional muscles in the **abdomen**.
- **Breathing Rate:** On average, a healthy human breathes **12–16 times/minute**.
- **Spirometer:** The volume of air involved in breathing movements can be estimated using a **spirometer**, which helps in the clinical assessment of pulmonary functions. (Note: Residual volume cannot be directly measured by simple spirometry because it is never expelled from the lungs).



Tutorial Diagram 2: Mechanism of breathing showing (a) Inspiration and (b) Expiration.

3. Respiratory Volumes, Capacities, and Partial Pressures

3.1 Respiratory Volumes

Respiratory Volume	Abbreviation	Standard Definition	Normal Average Value
Tidal Volume	TV	Volume of air inspired or expired during a normal respiration. (Minute volume: a healthy human inspires/expires approx. $6000 \text{ to } 8000\,\mathrm{mL/min}$).	$\approx 500\,\mathrm{mL}$
Inspiratory Reserve Volume	IRV	Additional volume of air a person can inspire by a forcible inspiration.	$2500 \text{ to } 3000\,\mathrm{mL}$
Expiratory Reserve Volume	ERV	Additional volume of air a person can expire by a forcible expiration.	$1000 \text{ to } 1100\,\mathrm{mL}$
Residual Volume	RV	Volume of air remaining in the lungs even after a forcible expiration.	$1100 \text{ to } 1200\,\mathrm{mL}$

3.2 Pulmonary Capacities (Derived for Clinical Diagnosis)

By adding specific respiratory volumes, various pulmonary capacities are derived for clinical assessment:

- **Inspiratory Capacity (IC):** Total volume of air a person can inspire after a normal expiration.
$$\mathrm{IC} = \mathrm{TV} + \mathrm{IRV} \approx 500 + (2500 \text{ to } 3000) = 3000 \text{ to } 3500\,\mathrm{mL}$$
- **Expiratory Capacity (EC):** Total volume of air a person can expire after a normal inspiration.
$$\mathrm{EC} = \mathrm{TV} + \mathrm{ERV} \approx 500 + (1000 \text{ to } 1100) = 1500 \text{ to } 1600\,\mathrm{mL}$$
- **Functional Residual Capacity (FRC):** Volume of air that will remain in the lungs after a normal expiration.
$$\mathrm{FRC} = \mathrm{ERV} + \mathrm{RV} \approx (1000 \text{ to } 1100) + (1100 \text{ to } 1200) = 2100 \text{ to } 2300\,\mathrm{mL}$$
- **Vital Capacity (VC):** Maximum volume of air a person can breathe in after a forced expiration, OR the maximum volume of air a person can breathe out after a forced inspiration.
$$\mathrm{VC} = \mathrm{ERV} + \mathrm{TV} + \mathrm{IRV} \approx 4000 \text{ to } 4600\,\mathrm{mL}$$
- **Total Lung Capacity (TLC):** Total volume of air accommodated in the lungs at the end of a forced inspiration.
$$\mathrm{TLC} = \mathrm{RV} + \mathrm{ERV} + \mathrm{TV} + \mathrm{IRV} = \mathrm{VC} + \mathrm{RV} \approx 5100 \text{ to } 5800\,\mathrm{mL}$$

3.3 Exchange of Gases and Partial Pressures

Alveoli are the primary sites of gas exchange; exchange also occurs between blood and tissues. $\mathrm{O_2}$ and $\mathrm{CO_2}$ are exchanged at these sites by **simple diffusion** mainly driven by **pressure/concentration gradients**.

The pressure contributed by an individual gas in a mixture of gases is called its **partial pressure**, represented as $\mathrm{pO_2}$ for oxygen and $\mathrm{pCO_2}$ for carbon dioxide.

Partial Pressures (in $\mathrm{mm\ Hg}$) of Oxygen and Carbon Dioxide at Different Parts Involved in Diffusion

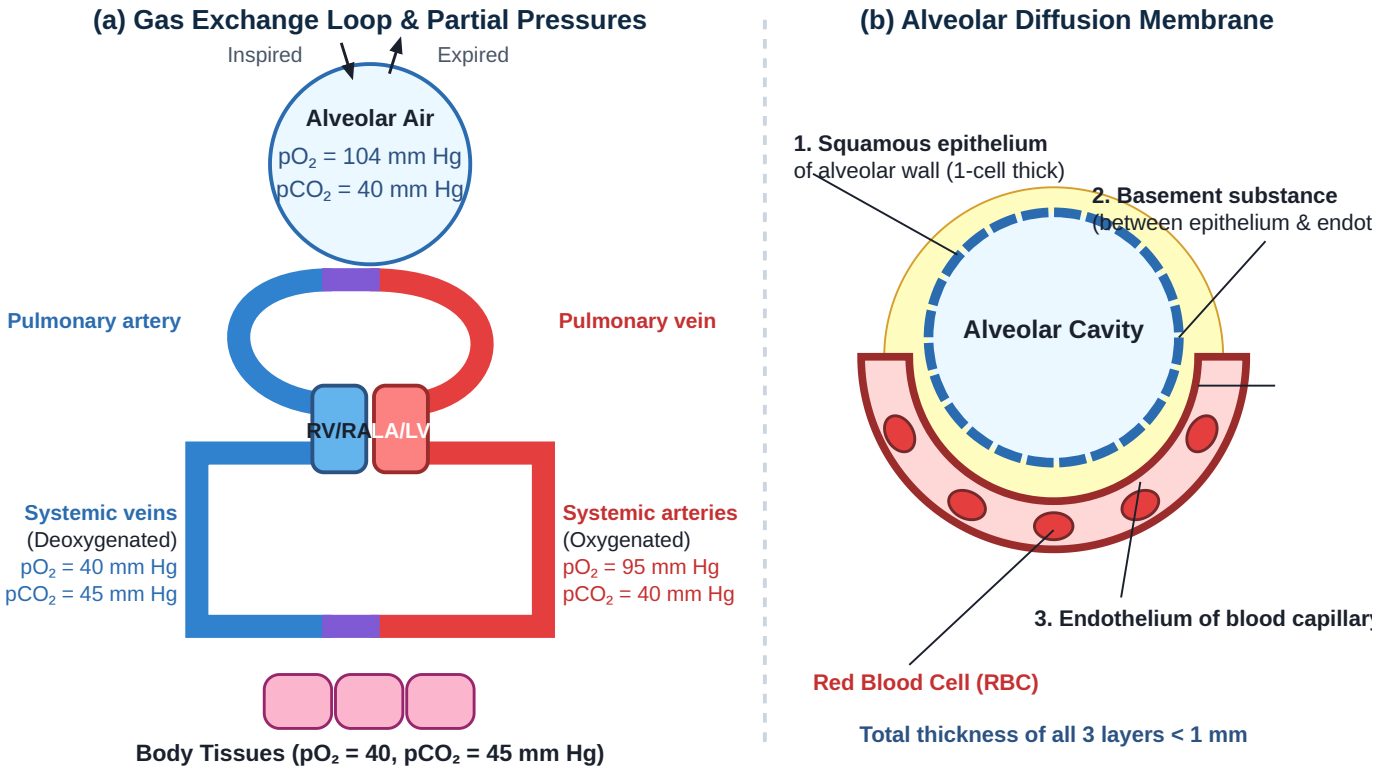
Respiratory Gas	Atmospheric Air	Alveoli	Blood (Deoxygenated)	Blood (Oxygenated)	Tissues
$\mathrm{O_2}$ ($\mathrm{pO_2}$)	159	104	40	95	40
$\mathrm{CO_2}$ ($\mathrm{pCO_2}$)	0.3	40	45	40	45

Factors Affecting Rate of Diffusion

1. **Partial Pressure / Concentration Gradient:**
   - For $\mathrm{O_2}$: Continuous gradient from Alveoli ($104\,\mathrm{mm\ Hg}$) → Deoxygenated blood ($40\,\mathrm{mm\ Hg}$), and from Oxygenated blood ($95\,\mathrm{mm\ Hg}$) → Tissues ($40\,\mathrm{mm\ Hg}$).

- For CO_2: Continuous gradient in the opposite direction, from Tissues ($45\ mm\ Hg$) → Blood ($40\ mm\ Hg$), and from Deoxygenated blood ($45\ mm\ Hg$) → Alveoli ($40\ mm\ Hg$).

2. **Solubility of the Gases:** The solubility of CO_2 is **20–25 times higher** than that of O_2. Consequently, the amount of CO_2 that can diffuse through the diffusion membrane per unit difference in partial pressure is much higher compared to that of O_2.
3. **Thickness of the Diffusion Membrane:** The alveolar-capillary diffusion membrane consists of **three major layers**, yet its total thickness is **much less than a millimetre**:
   - Thin **squamous epithelium of alveoli** (one-celled thick).
   - **Endothelium of alveolar capillaries** (single layer of endothelial cells).
   - **Basement substance** in between them (composed of a thin basement membrane supporting the alveolar squamous epithelium and the basement membrane surrounding the capillary endothelial cells).



Tutorial Diagram 3: (a) Exchange of gases at the alveolus and body tissues with partial pressures; (b) The 3-layered alveolar-capillary diffusion membrane.

4. Transport of Gases, Regulation, and Disorders

4.1 Quantitative Overview of Gas Transport in Blood

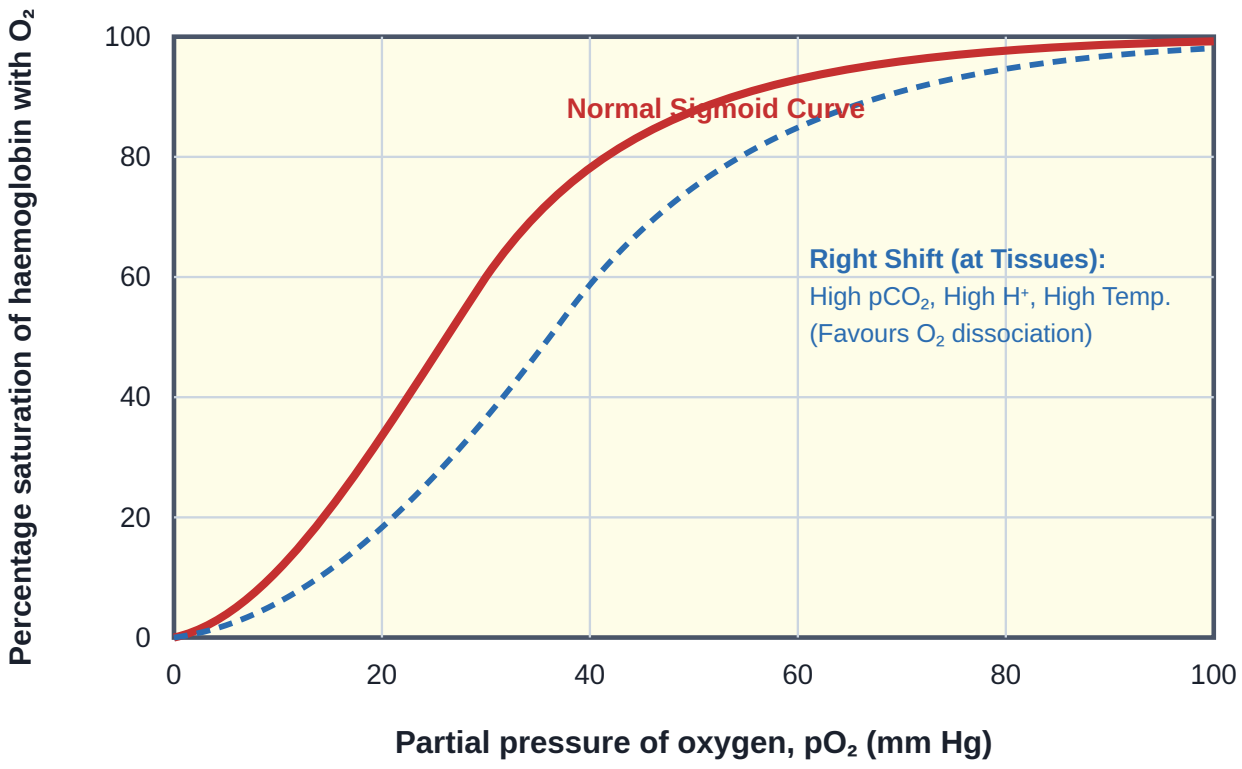
Blood is the medium of transport for both O_2 and CO_2:

Gas	Mode of Transport	Percentage Carried
Oxygen (O_2)	By RBCs (bound to haemoglobin as **oxyhaemoglobin**)	$\approx 97\%$
	Dissolved state through blood **plasma**	$\approx 3\%$
Carbon Dioxide (CO_2)	As **bicarbonate** (HCO_3^-)	$\approx 70\%$
	By RBCs (bound to haemoglobin as **carbamino-haemoglobin**)	$\approx 20–25\%$
	Dissolved state through blood **plasma**	$\approx 7\%$

4.2 Transport of Oxygen and the Oxygen Dissociation Curve

- **Haemoglobin (Hb):** A red-coloured, iron-containing pigment present in RBCs. $\mathrm{O_2}$ binds with haemoglobin in a **reversible manner** to form **oxyhaemoglobin**.
- **Stoichiometry:** Each haemoglobin molecule can carry a maximum of **four molecules of** $\mathrm{O_2}$.
- **Oxygen Dissociation Curve:** When percentage saturation of haemoglobin with $\mathrm{O_2}$ is plotted against $\mathrm{pO_2}$, a **sigmoid (S-shaped) curve** is obtained. Its sigmoidal pattern arises from cooperative binding (binding of the first $\mathrm{O_2}$ molecule facilitates the binding of subsequent $\mathrm{O_2}$ molecules to the remaining subunits of the haemoglobin tetramer).
- **Physiological Delivery Volume:** Every $100\ \mathrm{mL}$ of oxygenated blood can deliver around $5\ \mathrm{mL}$ **of** $\mathrm{O_2}$ to the tissues under normal physiological conditions.

Factor	Conditions in Alveoli (Favours $\mathrm{HbO_2}$ Formation / Association)	Conditions in Tissues (Favours $\mathrm{HbO_2}$ Dissociation / Right Shift)
Partial pressure of $\mathrm{O_2}$ ($\mathrm{pO_2}$)	High ($104\ \mathrm{mm\ Hg}$)	Low ($40\ \mathrm{mm\ Hg}$)
Partial pressure of $\mathrm{CO_2}$ ($\mathrm{pCO_2}$)	Low ($40\ \mathrm{mm\ Hg}$)	High ($45\ \mathrm{mm\ Hg}$)
$\mathrm{H^+}$ Concentration (Acidity / pH)	Lesser $\mathrm{H^+}$ concentration (Higher pH)	Higher $\mathrm{H^+}$ concentration (Lower pH)
Temperature	Lower temperature	Higher temperature



Tutorial Diagram 4: Oxygen dissociation curve showing the sigmoid relationship between partial pressure of oxygen and percentage saturation of haemoglobin.

4.3 Transport of Carbon Dioxide

1. **As Carbamino-haemoglobin ($20–25\%$):**
   - $\mathrm{CO_2}$ binds reversibly with haemoglobin to form **carbamino-haemoglobin**. This binding depends on $\mathrm{pCO_2}$, while $\mathrm{pO_2}$ is a major factor affecting it.

- **In tissues** (high $\mathrm{pCO_2}$, low $\mathrm{pO_2}$): More binding of $\mathrm{CO_2}$ with haemoglobin occurs.
- **In alveoli** (low $\mathrm{pCO_2}$, high $\mathrm{pO_2}$): Dissociation of $\mathrm{CO_2}$ from carbamino-haemoglobin takes place, delivering $\mathrm{CO_2}$ to the alveoli.

2. **As Bicarbonate Ions (70%):**
   - RBCs contain a very high concentration of the enzyme **carbonic anhydrase**, and minute quantities of the same enzyme are present in the plasma too.
   - This enzyme facilitates the following reversible reaction in both directions:

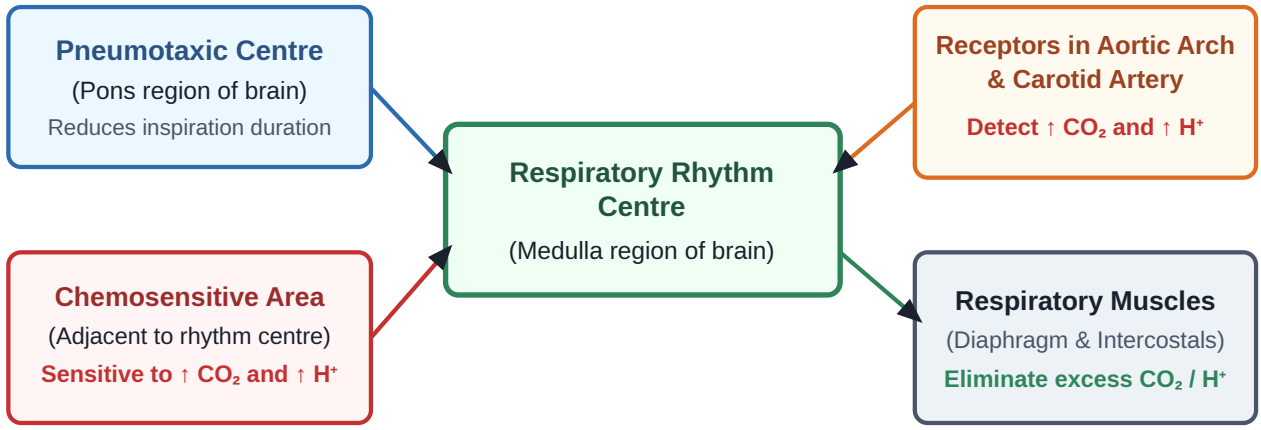
$$\mathrm{CO_2 + H_2O \xrightleftharpoons[\text{Carbonic anhydrase}]{\text{Carbonic anhydrase}} H_2CO_3 \xrightleftha}$$

- **At the tissue site:** Partial pressure of $\mathrm{CO_2}$ is high due to catabolism; $\mathrm{CO_2}$ diffuses into blood (RBCs and plasma) and reacts with water to form $\mathrm{HCO_3^-}$ and $\mathrm{H^+}$.
- **At the alveolar site:** $\mathrm{pCO_2}$ is low; the reaction proceeds in the opposite direction, forming $\mathrm{CO_2}$ and $\mathrm{H_2O}$. Thus, $\mathrm{CO_2}$ trapped as bicarbonate at the tissue level is transported to the alveoli and released out as gaseous $\mathrm{CO_2}$.
- **Physiological Delivery Volume:** Every $100\ \mathrm{mL}$ of deoxygenated blood delivers approximately $4\ \mathrm{mL}$ **of** $\mathrm{CO_2}$ to the alveoli.

4.4 Regulation of Respiration

Humans have a significant ability to maintain and moderate the respiratory rhythm to suit the demands of body tissues via the **neural system**:

- **Respiratory Rhythm Centre (Medulla):** A specialised centre present in the **medulla region** of the brain that is primarily responsible for regulating respiratory rhythm.
- **Pneumotaxic Centre (Pons):** Located in the **pons region** of the brain, it moderates the functions of the respiratory rhythm centre. Neural signals from this centre **reduce the duration of inspiration** and thereby alter (increase) the respiratory rate.
- **Central Chemosensitive Area (Adjacent to Rhythm Centre in Medulla):** Highly sensitive to $\mathrm{CO_2}$ **and hydrogen ions ($\mathrm{H^+}$)**. An increase in these substances activates this centre, which signals the respiratory rhythm centre to make adjustments so these substances can be eliminated.
- **Peripheral Chemoreceptors (Aortic Arch and Carotid Artery):** Receptors associated with the **aortic arch** and **carotid artery** also recognise changes in $\mathrm{CO_2}$ and $\mathrm{H^+}$ concentration and send necessary signals to the rhythm centre for remedial actions.
- **Role of Oxygen:** The role of oxygen ($\mathrm{O_2}$) in the regulation of respiratory rhythm is **quite insignificant**.



Tutorial Diagram 5: Neural and chemical regulation of human respiration. Note that O_2 plays a quite insignificant role in direct rhythm regulation.

4.5 Disorders of the Respiratory System

- **Asthma:** Difficulty in breathing causing **wheezing** due to **inflammation of bronchi and bronchioles**.
- **Emphysema:** A **chronic disorder** in which **alveolar walls are damaged**, due to which the **respiratory surface is decreased**. One of the major causes is **cigarette smoking**.
- **Occupational Respiratory Disorders:** Occur in industries involving **grinding or stone-breaking**, where so much dust is produced that the body's defense mechanism cannot fully cope. Long exposure causes **inflammation leading to fibrosis** (proliferation of fibrous tissues), resulting in serious lung damage. Workers in such industries should wear **protective masks**.
- **Hypoxia & High-Altitude Physiology: Hypoxia** is a condition of inadequate oxygen supply at the tissue level. When a person ascends a high hill (high altitude), atmospheric pressure and atmospheric pO_2 decrease, reducing alveolar pO_2 and causing arterial hypoxia. To compensate, the body initially increases breathing rate (hyperventilation) and heart rate, and over several days increases red blood cell (RBC) and haemoglobin production while decreasing the binding affinity of haemoglobin for O_2.

5. Comprehensive Worked Examples (Conceptual & Numerical)

Worked Example 1: Hourly Tidal Volume Calculation

Problem: Define Tidal Volume (TV). Calculate the approximate total Tidal Volume inspired or expired by a healthy human over the course of $1\,\mathrm{hour}$.

Solution:

- **Definition:** Tidal Volume (TV) is the volume of air inspired or expired during a normal, quiet respiration. For a healthy adult, $\mathrm{TV} \approx 500\,\mathrm{mL}$.
- **Step 1 (Minute Respiratory Volume):** A healthy human breathes 12 to 16 $\mathrm{times/minute}$.
$$\text{Volume per minute} = 500\,\mathrm{mL} \times (12 \text{ to } 16)\,\mathrm{min}^{-1} = 6000 \text{ to } 8000\,\mathrm{mL/min}$$
- **Step 2 (Hourly Tidal Volume):** Multiply the per-minute volume by 60 $\mathrm{minutes}$:
$$\text{Volume per hour (lower bound)} = 6000\,\mathrm{mL/min} \times 60\,\mathrm{min} = 360,000\,\mathrm{mL} = 360\,\mathrm{L}$$
$$\text{Volume per hour (upper bound)} = 8000\,\mathrm{mL/min} \times 60\,\mathrm{min} = 480,000\,\mathrm{mL} = 480\,\mathrm{L}$$
- **Conclusion:** A healthy human inspires or expires approximately $360,000\,\mathrm{mL}$ to $480,000\,\mathrm{mL}$ (or $360\,\mathrm{L}$ to $480\,\mathrm{L}$) of air per hour.

Worked Example 2: Deriving Pulmonary Capacities from Spirometric Data

Problem: A clinical spirometry report along with helium-dilution assessment for an adult subject gives the following measurements: Tidal Volume (TV) $= 500\,\mathrm{mL}$, Inspiratory Reserve Volume (IRV) $= 2800\,\mathrm{mL}$, Expiratory Reserve Volume (ERV) $= 1050\,\mathrm{mL}$, and Residual Volume (RV) $= 1150\,\mathrm{mL}$. Calculate: (a) Inspiratory Capacity (IC), (b) Functional Residual Capacity (FRC), (c) Vital Capacity (VC), and (d) Total Lung Capacity (TLC).

Solution:

- **(a) Inspiratory Capacity (IC):** Volume that can be inspired after a normal expiration:
$$\mathrm{IC} = \mathrm{TV} + \mathrm{IRV} = 500\,\mathrm{mL} + 2800\,\mathrm{mL} = 3300\,\mathrm{mL}$$
- **(b) Functional Residual Capacity (FRC):** Volume remaining in the lungs after a normal expiration:
$$\mathrm{FRC} = \mathrm{ERV} + \mathrm{RV} = 1050\,\mathrm{mL} + 1150\,\mathrm{mL} = 2200\,\mathrm{mL}$$
- **(c) Vital Capacity (VC):** Maximum volume of air breathed in after a forced expiration:
$$\mathrm{VC} = \mathrm{ERV} + \mathrm{TV} + \mathrm{IRV} = 1050\,\mathrm{mL} + 500\,\mathrm{mL} + 2800\,\mathrm{mL} = 4350\,\mathrm{mL}$$
- **(d) Total Lung Capacity (TLC):** Total air accommodated at the end of a forced inspiration:
$$\mathrm{TLC} = \mathrm{VC} + \mathrm{RV} = 4350\,\mathrm{mL} + 1150\,\mathrm{mL} = 5500\,\mathrm{mL}$$

Worked Example 3: Quantitative Gas Transport per Cardiac Output

Problem: Under normal resting physiological conditions, if the left ventricle pumps $5000\,\mathrm{mL}$ ($5\,\mathrm{L}$) of oxygenated blood per minute to systemic tissues and the right ventricle pumps $5000\,\mathrm{mL}$ of deoxygenated blood per minute to the lungs, calculate: (a) the volume of $\mathrm{O_2}$ delivered to the tissues per minute, and (b) the volume of $\mathrm{CO_2}$ delivered to the alveoli per minute.

Solution:

- **(a) $\mathrm{O_2}$ delivered to tissues:** Every $100\,\mathrm{mL}$ of oxygenated blood delivers around $5\,\mathrm{mL}$ of $\mathrm{O_2}$ to tissues under normal physiological conditions.
$$\text{Total } \mathrm{O_2} \text{ delivered per minute} = \frac{5\,\mathrm{mL}}{100\,\mathrm{mL}} \times 5000\,\mathrm{mL/min} = 250\,\mathrm{mL/min}$$
- **(b) $\mathrm{CO_2}$ delivered to alveoli:** Every $100\,\mathrm{mL}$ of deoxygenated blood delivers approximately $4\,\mathrm{mL}$ of $\mathrm{CO_2}$ to the alveoli.
$$\text{Total } \mathrm{CO_2} \text{ delivered per minute} = \frac{4\,\mathrm{mL}}{100\,\mathrm{mL}} \times 5000\,\mathrm{mL/min} = 200\,\mathrm{mL/min}$$

Worked Example 4: Anatomical Restriction of Gas Diffusion

Problem: Explain why diffusion of gases (O_2 and CO_2) occurs only in the alveolar region and not in the other parts of the human respiratory system.

Solution:

For rapid simple diffusion of gases between air and blood, the barrier must be extremely thin, moist, and richly supplied with blood capillaries. The **alveolar region** (alveoli and their ducts) forms the exchange part because its diffusion membrane—composed of one-cell-thick alveolar squamous epithelium, capillary endothelium, and the basement substance between them—has a total thickness of **much less than a millimetre** and is highly vascularised. In contrast, the **conducting part** (external nostrils up to terminal bronchioles) has thick walls (many supported by incomplete cartilaginous rings) and lacks the ultra-thin alveolar-capillary interface; its role is restricted to transporting, clearing, humidifying, and warming the air to body temperature.

6. NEET UG Exam-Aligned Practice Questions

Instructions: The following 15 original practice questions are designed strictly in accordance with the NEET UG examination standard, covering single-choice conceptual, numerical, multi-statement, matching, and assertion-reason formats.

Practice Question 1 (Single-Choice MCQ)

In the human respiratory system, the trachea is a straight tube extending up to the mid-thoracic cavity where it divides into right and left primary bronchi at the level of which vertebra, and which of the following structures lacks incomplete cartilaginous rings?

(A) 5th cervical vertebra; initial bronchioles

(B) 5th thoracic vertebra; terminal bronchioles and alveoli

(C) 7th thoracic vertebra; tertiary bronchi

(D) 2nd lumbar vertebra; secondary bronchi

Correct Answer: (B)

Worked Solution: The trachea extends up to the mid-thoracic cavity and divides at the level of the **5th thoracic vertebra** into right and left primary bronchi. Incomplete cartilaginous rings support the trachea, primary, secondary, and tertiary bronchi, and *initial bronchioles*, whereas **terminal bronchioles and alveoli** are very thin-walled and lack cartilaginous rings.

Practice Question 2 (Single-Choice MCQ)

What will be the partial pressures of oxygen (pO_2) and carbon dioxide (pCO_2) in atmospheric air compared to those in alveolar air?

(A) pO_2 lesser, pCO_2 higher

(B) pO_2 higher, pCO_2 lesser

(C) pO_2 higher, pCO_2 higher

(D) pO_2 lesser, pCO_2 lesser

Correct Answer: (B)

Worked Solution: In atmospheric air, $pO_2 = 159\,\text{mm Hg}$ and $pCO_2 = 0.3\,\text{mm Hg}$. In alveolar air, $pO_2 = 104\,\text{mm Hg}$ and $pCO_2 = 40\,\text{mm Hg}$. Therefore, in atmospheric air compared to alveolar air, pO_2 is higher ($159 > 104$) and pCO_2 is lesser ($0.3 < 40$).

Practice Question 3 (Single-Choice MCQ)

During normal inspiration in a healthy human, which of the following muscular contractions and volume changes occur?

(A) Diaphragm contracts to increase thoracic volume in the dorso-ventral axis; internal intercostals contract to increase thoracic volume in the antero-posterior axis.

(B) Diaphragm contracts to increase thoracic volume in the antero-posterior axis; external intercostals contract to increase thoracic volume in the dorso-ventral axis.

(C) Diaphragm relaxes and arches upwards; intra-pulmonary pressure becomes higher than atmospheric pressure.

(D) External intercostal muscles relax; intra-pulmonary pressure drops below atmospheric pressure.

Correct Answer: (B)

Worked Solution: Inspiration is initiated by the contraction of the **diaphragm**, which increases the volume of the thoracic chamber in the **antero-posterior axis**. Simultaneously, contraction of the **external intercostal muscles** lifts up the ribs and sternum, causing an increase in the volume of the thoracic chamber in the **dorso-ventral axis**. This overall increase in thoracic and pulmonary volume decreases intra-pulmonary pressure below atmospheric pressure.

Practice Question 4 (Single-Choice Numerical MCQ)

What is the volume of air that remains in the lungs after a normal expiration, and what is its standard normal value in a healthy adult human?

(A) Residual Volume (RV); 1100 to $1200\,\mathrm{mL}$

(B) Expiratory Capacity (EC); 1500 to $1600\,\mathrm{mL}$

(C) Functional Residual Capacity (FRC); 2100 to $2300\,\mathrm{mL}$

(D) Vital Capacity (VC); 4000 to $4600\,\mathrm{mL}$

Correct Answer: (C)

Worked Solution: The volume of air that remains in the lungs after a *normal* expiration is the **Functional Residual Capacity** (FRC), which equals Expiratory Reserve Volume plus Residual Volume ($\mathrm{ERV} + \mathrm{RV}$). Using standard values: $\mathrm{ERV} = 1000$ to $1100\,\mathrm{mL}$ and $\mathrm{RV} = 1100$ to $1200\,\mathrm{mL}$, giving $\mathrm{FRC} = 2100$ to $2300\,\mathrm{mL}$. (Note: Residual Volume alone is the air remaining after a *forcible* expiration).

Practice Question 5 (Match-the-Column MCQ)

Match the animal groups in Column I with their respective primary respiratory structures/modes in Column II:

Column I (Animal Group)	Column II (Respiratory Structure)
P. Coelenterates and Flatworms	1. Moist cuticle
Q. Earthworms	2. Network of tracheal tubes
R. Insects	3. Entire body surface by simple diffusion
S. Aquatic arthropods and Molluscs	4. Special vascularised structures called gills

(A) P–3, Q–1, R–2, S–4

(B) P–1, Q–3, R–4, S–2

(C) P–3, Q–2, R–1, S–4

(D) P–4, Q–1, R–2, S–3

Correct Answer: (A)

Worked Solution: Lower invertebrates like sponges, coelenterates, and flatworms exchange gases by simple diffusion over their entire body surface (P–3). Earthworms use their moist cuticle (Q–1). Insects have a network of tubes called tracheal tubes (R–2). Aquatic arthropods and molluscs use special vascularised structures called gills for branchial respiration (S–4).

Practice Question 6 (Multi-Statement Evaluation)

Read the following four statements (I–IV) regarding the transport of gases in human blood:

I. About 97% of O_2 is transported by RBCs in the blood, whereas the remaining 3% is carried in a dissolved state through the plasma.

II. Nearly $20–25\%$ of CO_2 is carried in a dissolved state through plasma, while 7% is transported by RBCs as carbamino-haemoglobin.

III. Every $100\ mL$ of oxygenated blood delivers around $5\ mL$ of O_2 to the tissues under normal physiological conditions.

IV. Every $100\ mL$ of deoxygenated blood delivers approximately $4\ mL$ of CO_2 to the alveoli.

Which of the above statements are **correct**?

(A) I, II, and III only

(B) I, III, and IV only

(C) II and IV only

(D) I, II, III, and IV

Correct Answer: (B)

Worked Solution: Statements I, III, and IV are completely accurate. Statement II is incorrect because nearly $20–25\%$ of CO_2 is transported by RBCs (as carbamino-haemoglobin), 70% is carried as bicarbonate, and only about 7% of CO_2 is carried in a dissolved state through plasma.

Practice Question 7 (Single-Choice MCQ)

Which of the following sets of physiological conditions is most favourable for the **dissociation** of oxygen from oxyhaemoglobin?

(A) High pO_2, low pCO_2, lesser H^+ concentration, and lower temperature

(B) Low pO_2, low pCO_2, lesser H^+ concentration, and lower temperature

(C) Low pO_2, high pCO_2, high H^+ concentration, and higher temperature

(D) High pO_2, high pCO_2, lesser H^+ concentration, and higher temperature

Correct Answer: (C)

Worked Solution: In the metabolically active tissues, **low** pO_2, **high** pCO_2, **high** H^+ **concentration** (lower pH), and **higher temperature** exist. All four of these conditions shift the oxygen dissociation curve to the right and favour the dissociation of oxygen from oxyhaemoglobin. Conversely, Option (A) describes the conditions in the alveoli that favour oxyhaemoglobin formation.

Practice Question 8 (Assertion–Reason)

Assertion (A): The amount of CO_2 that can diffuse through the alveolar diffusion membrane per unit difference in partial pressure is much higher compared to that of O_2.

Reason (R): The solubility of CO_2 is $20–25$ times higher than that of O_2.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Answer: (A)

Worked Solution: Both statements are true and directly linked. Even though the partial pressure gradient for CO_2 ($45\ mm\ Hg$ to $40\ mm\ Hg$, a difference of only $5\ mm\ Hg$) is much smaller than that for O_2 ($104\ mm\ Hg$ to $40\ mm\ Hg$), the solubility of CO_2 is $20–25$ times higher than that of O_2. Therefore, the amount of CO_2 diffusing per unit partial pressure difference is much greater.

Practice Question 9 (Single-Choice MCQ)

Examine the partial pressures of O_2 and CO_2 across different parts of the human body. Which of the following pairs of sites have identical values for both pO_2 and pCO_2?

(A) Alveoli and Oxygenated blood

(B) Deoxygenated blood and Tissues

(C) Oxygenated blood and Tissues

(D) Alveoli and Deoxygenated blood

Correct Answer: (B)

Worked Solution: Referring to standard physiological partial pressures: in **Deoxygenated blood**, $pO_2 = 40\ mm\ Hg$ and $pCO_2 = 45\ mm\ Hg$. In **Tissues**, $pO_2 = 40\ mm\ Hg$ and $pCO_2 = 45\ mm\ Hg$. Thus, deoxygenated blood and tissues share the exact same partial pressures ($40\ mm\ Hg$ and $45\ mm\ Hg$). (In contrast, Alveoli have $pO_2 = 104\ mm\ Hg$, whereas Oxygenated blood has $pO_2 = 95\ mm\ Hg$).

Practice Question 10 (Single-Choice MCQ)

Which of the following accurately describes the distribution and role of the enzyme **carbonic anhydrase** in human blood?

(A) It is present in high concentration in blood plasma and absent in RBCs, catalysing only the formation of HCO_3^- at the tissue level.

(B) It is present in very high concentration in RBCs and in minute quantities in plasma, facilitating the reversible conversion between $CO_2 + H_2O$, H_2CO_3, and $HCO_3^- + H^+$ in both directions.

(C) It is located exclusively on the alveolar squamous epithelium to split CO_2 into carbon and O_2.

(D) It directly binds CO_2 to the iron atom of haemoglobin to form carbamino-haemoglobin.

Correct Answer: (B)

Worked Solution: RBCs contain a very high concentration of the enzyme carbonic anhydrase, and minute quantities of the same are present in the plasma too. It catalyses the reversible reaction $CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons HCO_3^- + H^+$ in both directions: forming HCO_3^- and H^+ at the tissues (where pCO_2 is high) and forming CO_2 and H_2O at the alveoli (where pCO_2 is low).

Practice Question 11 (Assertion–Reason)

Assertion (A): Neural signals from the pneumotaxic centre in the pons region of the brain can increase the respiratory rate.

Reason (R): Neural signals from the pneumotaxic centre reduce the duration of inspiration, thereby moderating the functions of the respiratory rhythm centre.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Answer: (A)

Worked Solution: The pneumotaxic centre in the pons region moderates the functions of the medullary respiratory rhythm centre. By sending neural signals that **reduce the duration of inspiration**, each breathing cycle is shortened, which consequently increases the respiratory rate.

Practice Question 12 (Single-Choice MCQ)

Regarding the chemical regulation of respiration in humans, which of the following statements is **correct**?

(A) Receptors associated with the pulmonary vein and vena cava are the primary sensors of blood O_2 levels.

(B) A chemosensitive area adjacent to the rhythm centre and receptors associated with the aortic arch and carotid artery recognise increases in CO_2 and H^+ concentration, whereas the role of oxygen in regulating respiratory rhythm is quite insignificant.

(C) A decrease in CO_2 and H^+ concentration strongly activates the medullary chemosensitive area to increase ventilation.

(D) Oxygen concentration is the primary chemical stimulus activating the central chemosensitive area in the medulla.

Correct Answer: (B)

Worked Solution: Both the chemosensitive area (adjacent to the medullary rhythm centre) and peripheral receptors (associated with the **aortic arch and carotid artery**) are highly sensitive to increases in CO_2 **and** H^+ concentration. They signal the rhythm centre to make adjustments to eliminate excess CO_2 and H^+. The role of oxygen in the regulation of respiratory rhythm is quite insignificant.

Practice Question 13 (Multi-Statement Evaluation)

Consider the following statements regarding the alveolar-capillary diffusion membrane:

I. It is composed of three major layers: thin squamous epithelium of alveoli, endothelium of alveolar capillaries, and the basement substance between them.

II. The basement substance is composed of a thin basement membrane supporting the squamous epithelium and the basement membrane surrounding the single-layer endothelial cells of capillaries.

III. The total thickness of the diffusion membrane is approximately $2.5\ \text{mm}$.

Which of the statements given above is/are **correct**?

(A) I only

(B) I and II only

(C) II and III only

(D) I, II, and III

Correct Answer: (B)

Worked Solution: Statements I and II accurately describe the three layers and the composition of the basement substance. Statement III is incorrect because the total thickness of the diffusion membrane is **much less than a millimetre** ($< 1\ \text{mm}$), which facilitates rapid gas diffusion.

Practice Question 14 (Single-Choice MCQ)

Match the respiratory disorder with its characteristic pathological feature:

(A) **Asthma:** Chronic damage to alveolar walls causing decreased respiratory surface, mainly due to cigarette smoking.

(B) **Emphysema:** Proliferation of fibrous tissues (fibrosis) in the lungs due to long exposure to stone-breaking dust.

(C) **Emphysema:** Chronic disorder in which alveolar walls are damaged, decreasing the respiratory surface, with cigarette smoking as a major cause.

(D) **Occupational Respiratory Disorder:** Wheezing caused exclusively by reversible spasm of the epiglottis.

Correct Answer: (C)

Worked Solution: Emphysema is a chronic disorder in which alveolar walls are damaged, decreasing the respiratory surface; cigarette smoking is one of its major causes. **Asthma** is difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles. **Occupational respiratory disorders** involve inflammation leading to **fibrosis** (proliferation of fibrous tissues) from long-term dust exposure in grinding or stone-breaking industries.

Practice Question 15 (Single-Choice Numerical/Conceptual MCQ)

For a healthy adult human, which of the following mathematical relationships among respiratory volumes and capacities is **incorrect**?

(A) $\mathrm{TLC} = \mathrm{VC} + \mathrm{RV}$

(B) $\mathrm{VC} = \mathrm{IC} + \mathrm{ERV}$

(C) $\mathrm{TLC} = \mathrm{IC} + \mathrm{FRC}$

(D) $\mathrm{FRC} = \mathrm{TV} + \mathrm{ERV}$

Correct Answer: (D)

Worked Solution: Let us verify each relation using the fundamental volume definitions:
• (A) $\mathrm{TLC} = \mathrm{RV} + \mathrm{ERV} + \mathrm{TV} + \mathrm{IRV} = \mathrm{VC} + \mathrm{RV}$ (Correct).
• (B) Since $\mathrm{IC} = \mathrm{TV} + \mathrm{IRV}$, $\mathrm{IC} + \mathrm{ERV} = \mathrm{TV} + \mathrm{IRV} + \mathrm{ERV} = \mathrm{VC}$ (Correct).
• (C) Since $\mathrm{IC} = \mathrm{TV} + \mathrm{IRV}$ and $\mathrm{FRC} = \mathrm{ERV} + \mathrm{RV}$, their sum is $\mathrm{TV} + \mathrm{IRV} + \mathrm{ERV} + \mathrm{RV} = \mathrm{TLC}$ (Correct).
• (D) $\mathrm{TV} + \mathrm{ERV}$ is **Expiratory Capacity** (EC), whereas **Functional Residual Capacity** (FRC) is $\mathrm{ERV} + \mathrm{RV}$. Hence, (D) is incorrect.

7. Common NEET Pitfalls and Revision Checklist

Common Exam Pitfalls to Avoid

- **Pitfall 1 — Cartilaginous Rings in Bronchioles:** Students often assume *all* bronchioles lack cartilaginous rings or *all* have them. Remember: **Initial bronchioles** have incomplete cartilaginous rings, whereas **terminal bronchioles** do not.
- **Pitfall 2 — Normal Expiration vs. Forced Expiration Volume:** Air remaining in the lungs after a *normal* expiration is FRC ($\mathrm{ERV} + \mathrm{RV}$), whereas air remaining after a *forcible* expiration is RV.
- **Pitfall 3 — Dissolved $\mathrm{O_2}$ vs. Dissolved $\mathrm{CO_2}$ in Plasma:** 3% of $\mathrm{O_2}$ is carried dissolved in plasma, whereas 7% of $\mathrm{CO_2}$ is carried dissolved in plasma. Do not swap 7% (dissolved $\mathrm{CO_2}$) with 20–25% (carbamino-haemoglobin).
- **Pitfall 4 — Alveolar $\mathrm{pO_2}$ vs. Oxygenated Blood $\mathrm{pO_2}$:** Alveolar $\mathrm{pO_2}$ is $104\,\mathrm{mm\,Hg}$, while oxygenated systemic arterial blood $\mathrm{pO_2}$ is $95\,\mathrm{mm\,Hg}$ (not $104\,\mathrm{mm\,Hg}$). However, $\mathrm{pCO_2}$ is $40\,\mathrm{mm\,Hg}$ in both alveoli and oxygenated blood.
- **Pitfall 5 — Role of $\mathrm{O_2}$ in Respiratory Rhythm:** A frequent trap is stating that falling $\mathrm{O_2}$ directly stimulates the medullary chemosensitive area. The central chemosensitive area and aortic/carotid receptors respond to $\mathrm{CO_2}$ **and** $\mathrm{H^+}$; the role of $\mathrm{O_2}$ in respiratory rhythm regulation is **quite insignificant**.

High-Yield Revision Checklist

- Trachea bifurcates at the **5th thoracic vertebra** into right and left primary bronchi.
- Contraction of diaphragm increases thoracic volume in the **antero-posterior axis**; contraction of external intercostals increases thoracic volume in the **dorso-ventral axis**.
- Healthy breathing rate: **12–16 times/minute**; Minute volume: 6000–$8000\,\mathrm{mL/min}$.
- Standard Volumes: $\mathrm{TV} = 500\,\mathrm{mL}$, $\mathrm{IRV} = 2500$–$3000\,\mathrm{mL}$, $\mathrm{ERV} = 1000$–$1100\,\mathrm{mL}$, $\mathrm{RV} = 1100$–$1200\,\mathrm{mL}$.
- $\mathrm{CO_2}$ solubility is **20–25 times higher** than $\mathrm{O_2}$ solubility; diffusion membrane has **3 layers** with total thickness $< 1\,\mathrm{mm}$.
- $1\,\mathrm{Hb}$ molecule carries a maximum of $4\,\mathrm{O_2}$ molecules; $100\,\mathrm{mL}$ oxygenated blood delivers $5\,\mathrm{mL}\,\mathrm{O_2}$ to tissues; $100\,\mathrm{mL}$ deoxygenated blood delivers $4\,\mathrm{mL}\,\mathrm{CO_2}$ to alveoli.
- Medulla = **Respiratory rhythm centre** & **Chemosensitive area**; Pons = **Pneumotaxic centre**.

8. Concise Final Recap

Respiration integrates pulmonary ventilation, alveolar diffusion, blood gas transport, tissue diffusion, and cellular catabolism. In humans, atmospheric air passes through the conducting zone (nostrils to terminal bronchioles) to reach the exchange zone (alveoli). Breathing is driven by negative and positive intra-pulmonary pressure gradients created by the diaphragm and intercostal muscles. Gas exchange across the sub-millimetre, three-layered alveolar-capillary membrane follows partial pressure gradients ($\mathrm{pO_2}$: $104 \rightarrow 40\,\mathrm{mm\,Hg}$; $\mathrm{pCO_2}$: $45 \rightarrow 40\,\mathrm{mm\,Hg}$). Blood transports 97% of $\mathrm{O_2}$ as oxyhaemoglobin (displaying a cooperative sigmoidal dissociation curve sensitive to $\mathrm{pCO_2}$, $\mathrm{H^+}$, and temperature) and 70% of $\mathrm{CO_2}$ as bicarbonate via carbonic anhydrase (along with 20–25% as carbamino-haemoglobin and 7% dissolved in plasma). Finally, the medullary respiratory rhythm centre, modulated by the pontine pneumotaxic centre and $\mathrm{CO_2}/\mathrm{H^+}$-sensitive central and peripheral chemoreceptors, adjusts ventilation to match metabolic demand.